

Group 7: The halogens (extension material)

Group 7 is another group in the Periodic Table, known as the **halogens**. The group includes fluorine, chlorine and bromine. These elements have a number of properties in common. The first two elements (fluorine and chlorine) are gases at room temperature. Bromine is a liquid. The most reactive is fluorine, then chlorine. Bromine is the least reactive of the three.

Group 7 (halogens)

Element	Atomic number	Electronic structure	Mass number	Colour	Melting point in °C	Boiling point in °C
fluorine, F	9	2,7	19	pale yellow	-220	-188
chlorine, Cl	17	2,8,7	35	yellowish green	-101	-34
bromine, Br	35		80	brown	-7	59

Questions

- 9 Are the halogens metals or non-metals?
- 10 What is the trend in melting points in Group 7?
- 11 What is the trend in colour in Group 7?
- 12 What is the trend in boiling points in Group 7?
- 13 What would you predict about the boiling point and melting point of iodine, the next element in Group 7?
- 14 Would you expect iodine to be more or less reactive than bromine?
- 15 The electronic structure for bromine is missing from the table. How many electrons do you predict will be in the outer shell?

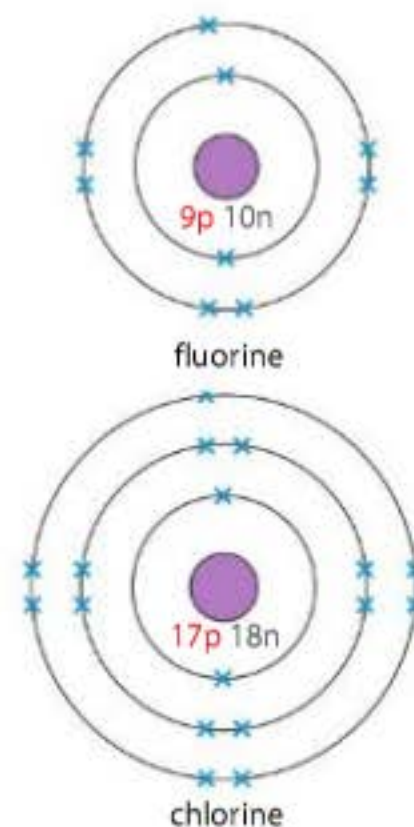
The structure of fluorine and chlorine

Fluorine has an atomic number of 9 and a mass number of 19. This atom contains 9 protons, 9 electrons and 10 neutrons. The electronic structure is 2,7. This means that the first electron shell has 2 electrons and is full. The second electron shell has seven electrons.

Chlorine has an atomic number of 17 and a mass number of 35. This atom contains 17 protons, 17 electrons and 18 neutrons. The electronic structure is 2,8,7. This means that the first electron shell has two electrons and is full. The second electron shell has eight electrons and is full. The third electron shell has seven electrons.

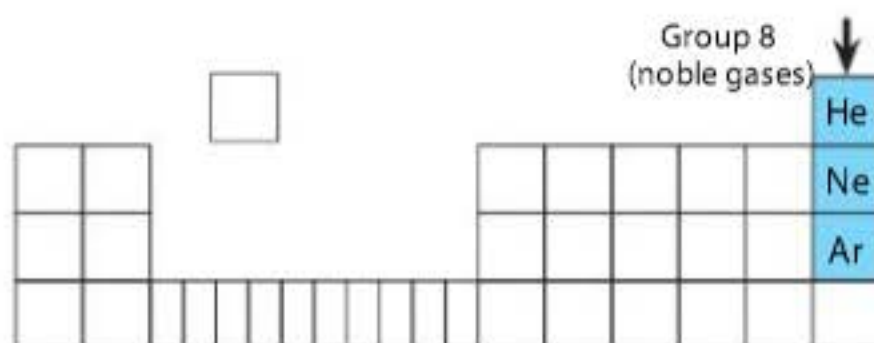
Questions

- 16 What happens to the size of the atoms as you go down this group?
- 17 What similarity is there in the structure of these atoms?
- 18 Suggest why this group is called Group 7.



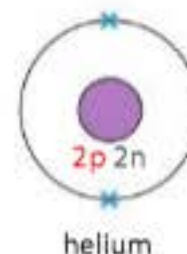
Group 8: The noble gases

Group 8 includes the elements helium, neon and argon. They are all gases. They are inert (unreactive) and do not form compounds. They are called the **noble gases**.

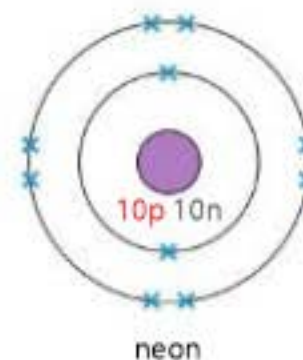


Element	Atomic number	Electronic structure	Mass number	Melting point in °C	Boiling point in °C
helium, He	2	2	4	-270	-269
neon, Ne	10	2,8	20	-249	-246
argon, Ar	18	2,8,8	40	-189	-186

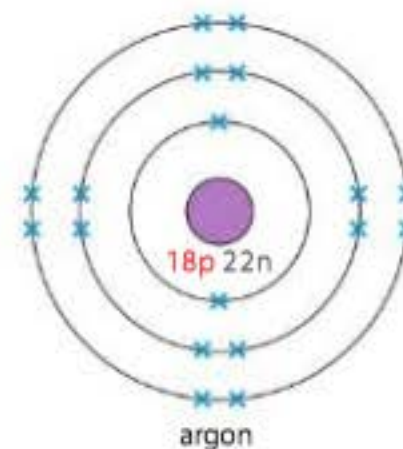
Helium has an atomic number of 2 and a mass number of 4. This atom contains 2 protons, 2 electrons and 2 neutrons. The electrons are arranged with 2 in the first shell. The shell is full.



Neon has an atomic number of 10 and a mass number of 20. This atom contains 10 protons, 10 electrons and 10 neutrons. The electrons are arranged with 2 in the first shell and 8 in the second shell. Both shells are full.



Argon has an atomic number of 18 and a mass number of 40. This atom contains 18 protons, 18 electrons and 22 neutrons. The electrons are arranged with 2 in the first shell, 8 in the second shell and 8 in the third shell. All three shells are full.



Questions

- 19 What trend in melting points can be seen in Group 8?
- 20 What happens to the size of the atoms as you go down Group 8?
- 21 What similarity is there in the structure of these atoms?
- 22 Suggest why this group is called Group 8.
- 23 What predictions can you make about the melting point and boiling point of krypton, which is the next gas in this group?

How does the atomic structure of an element relate to its position in the Periodic Table?

Summary checklist

- ☐ I can explain how the structure of the Periodic Table is related to the atomic structure of the elements.
- ☐ I can identify the similarities and differences between the atomic structure of elements in the same group in the Periodic Table.
- ☐ I can use the Periodic Table to predict the structure and properties of elements.
- ☐ I can make accurate observations.